
MODEL SELECTION AND VALIDATION REPORT (2012_2016_biomass model)

OBSERVATIONS:794

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1_with_anisotropy

MODEL SUMMARY (2012_2016_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.12	0.18
kde_fric	0.26	0.04
kde_feve	-0.28	0.04
tric	0.06	0.04
teve	-0.46	0.04
sbt_mean	0.26	0.08
sbt_sd	0.07	0.05
sst_sd	0.09	0.05
sic_mean	-0.13	0.08
log_chla_mean	0.00	0.04
depth	0.06	0.05

Dispersion parameter: 0.75

Spatiotemporal AR1 correlation (rho): 0.81

Matérn anisotropic range (spatiotemporal): 179.8 to 731.4 at 20 deg.

Spatiotemporal marginal AR1 SD: 0.52

REML criterion at convergence: 6410.129

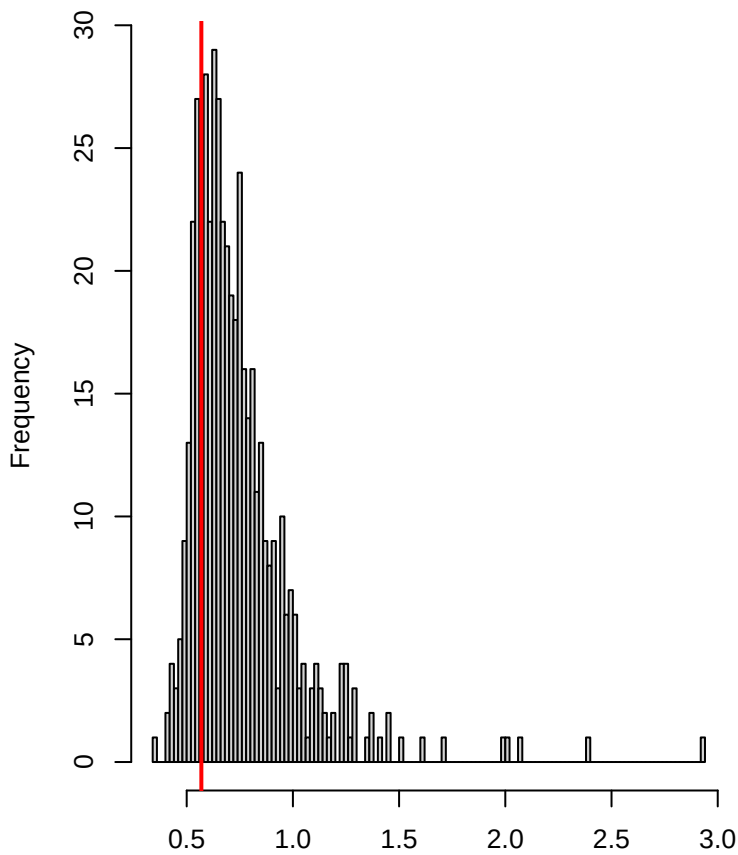
See ?tidy.sdmTMB to extract these values as a data frame.

See ?plot_anisotropy to plot the anisotropic range.

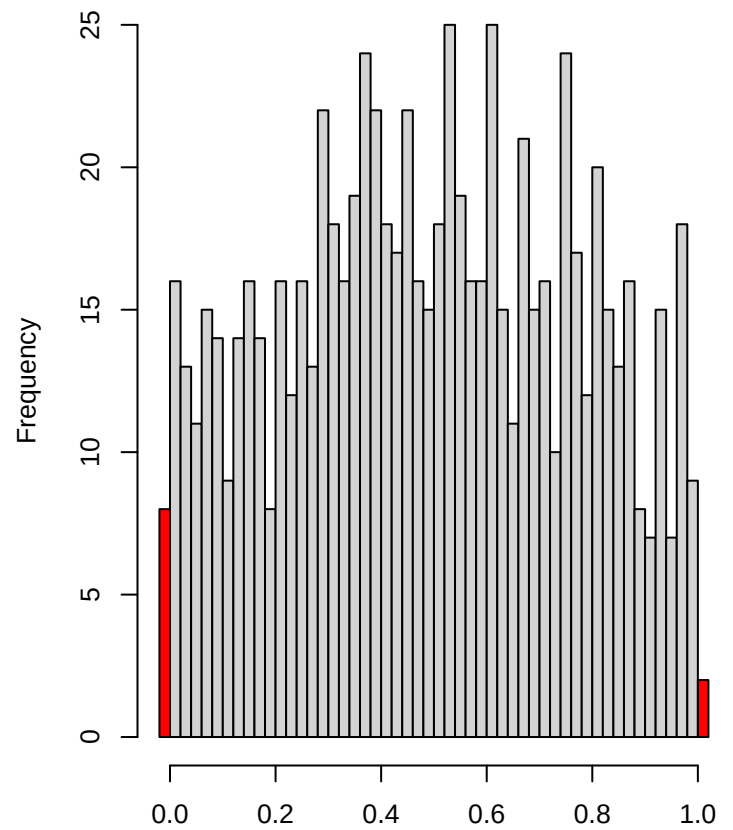
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy	17	12854.26	2.1480061
spatiotemporal_random_fields_ar1	15	12856.41	12.8043375
spatial_random_fields_with_anisotropy	16	12869.21	0.4723002
spatial_random_fields	14	12869.68	13.5630827
spatiotemporal_random_fields_iid_with_anisotropy	16	12883.25	2.9583162
spatiotemporal_random_fields_iid	14	12886.20	84.1586967
error_only	12	12970.36	NA

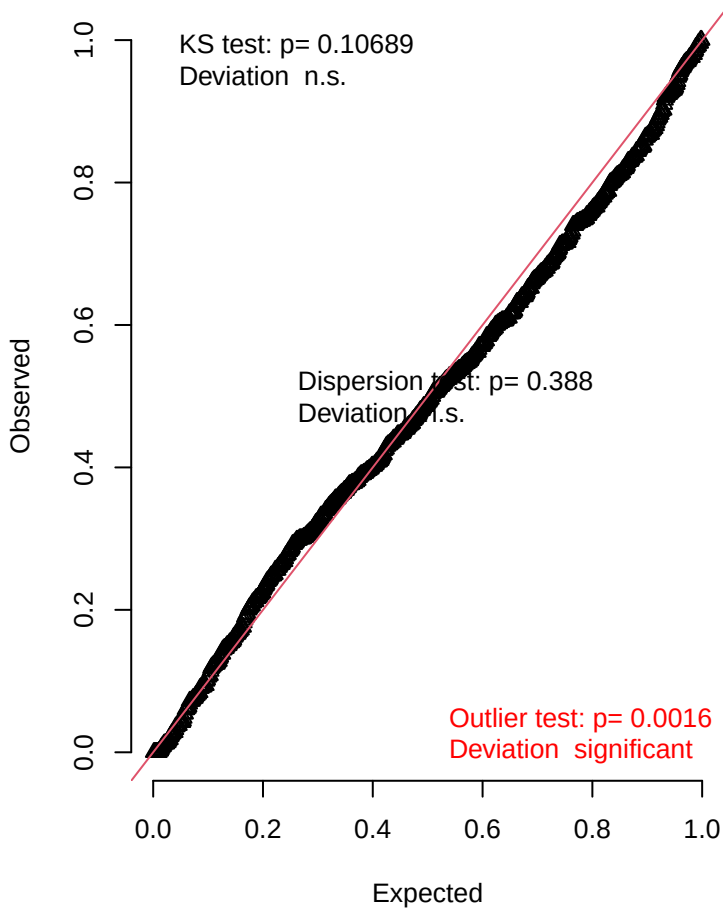
DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



Outlier test significant

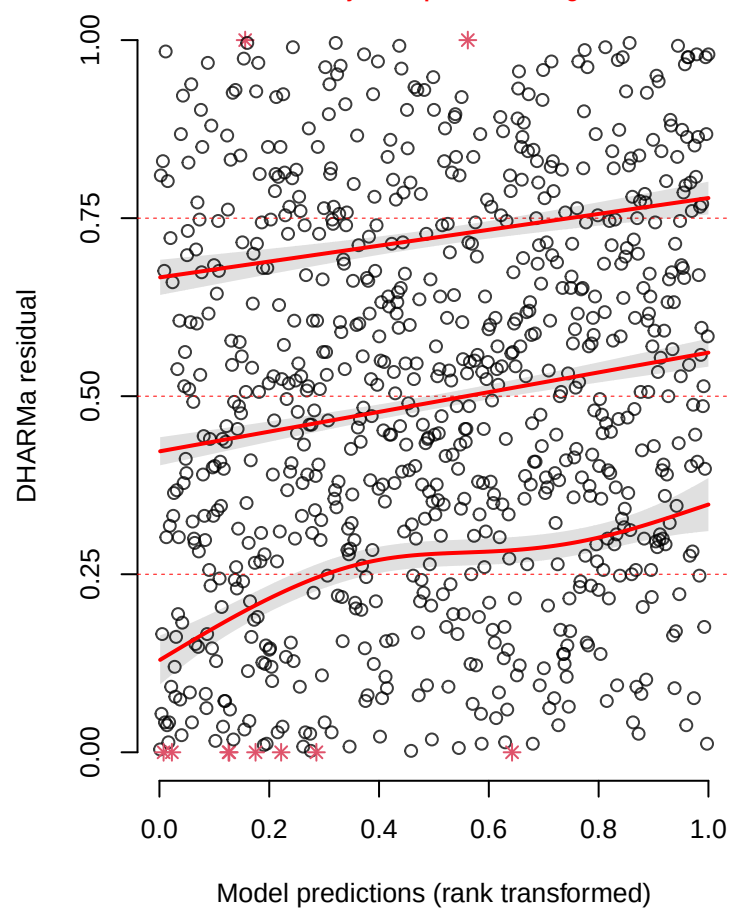


QQ plot residuals

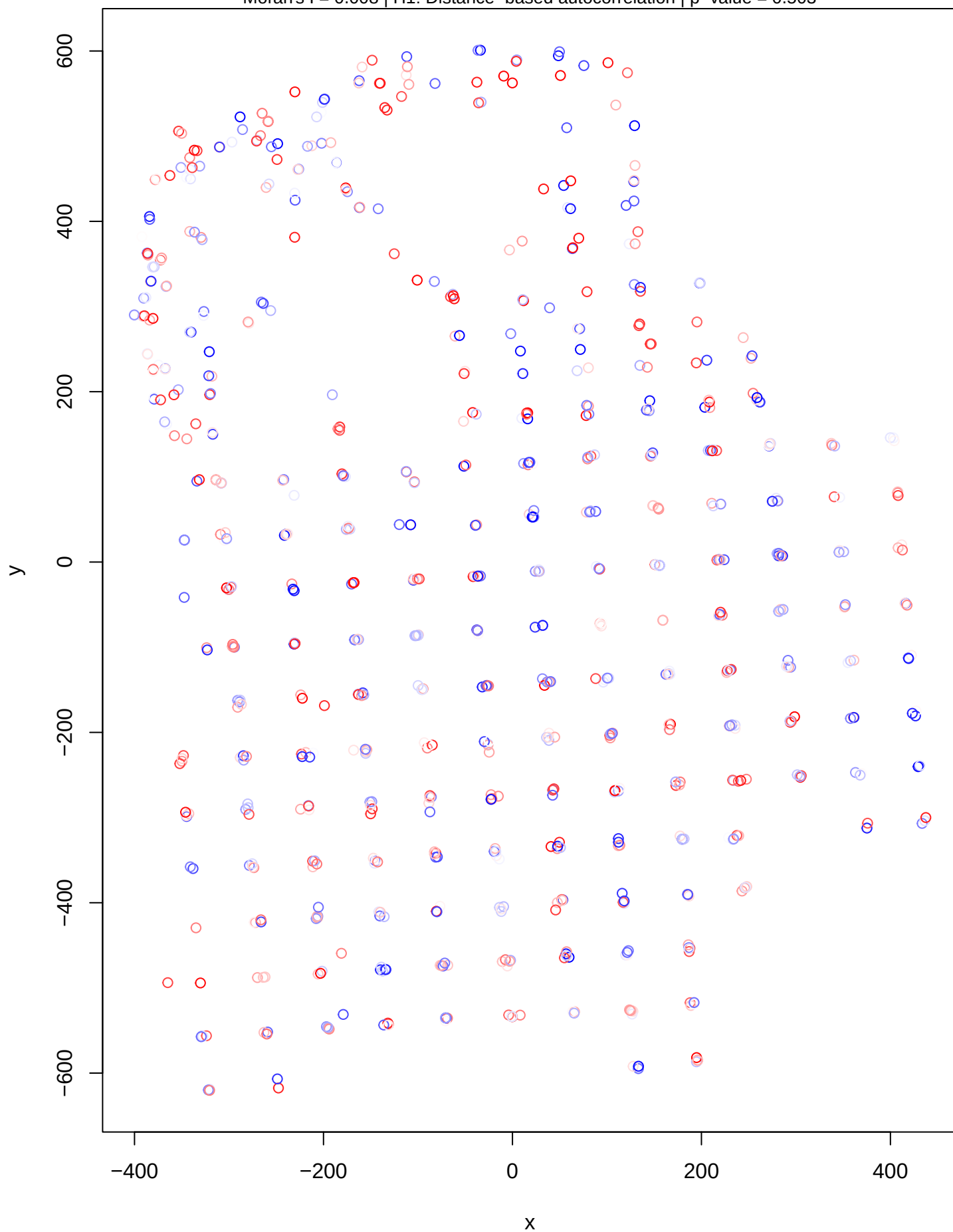


Residuals (outliers are marked red)

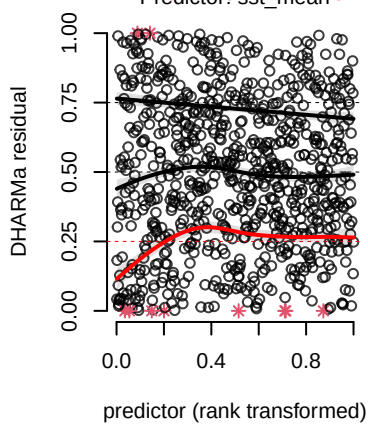
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



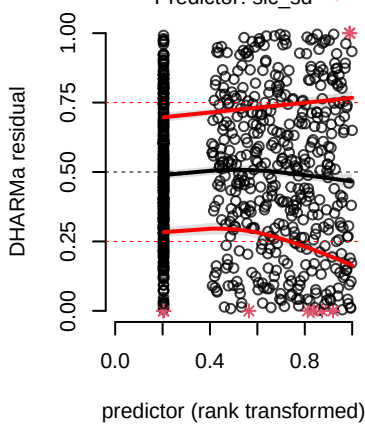
DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.008 | H1: Distance-based autocorrelation | p-value = 0.503



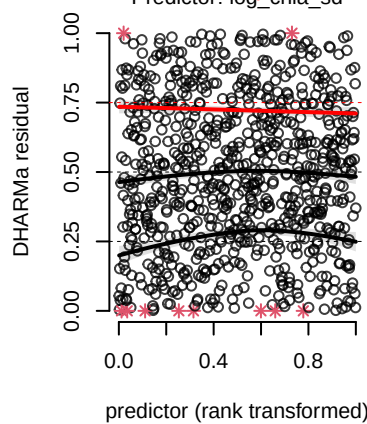
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



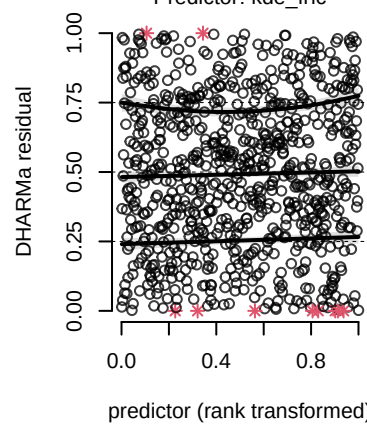
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



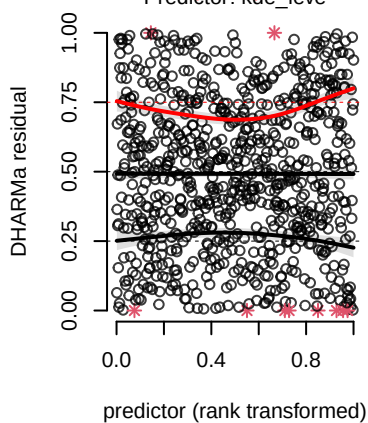
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



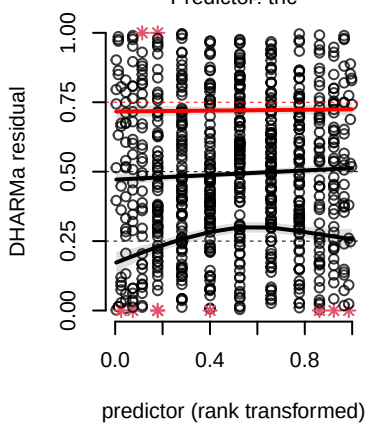
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: kde_fric



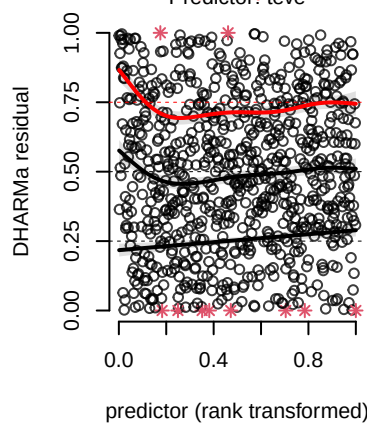
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



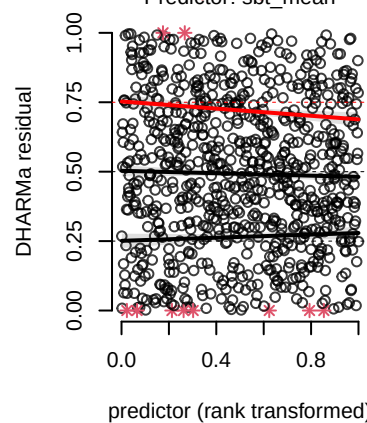
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



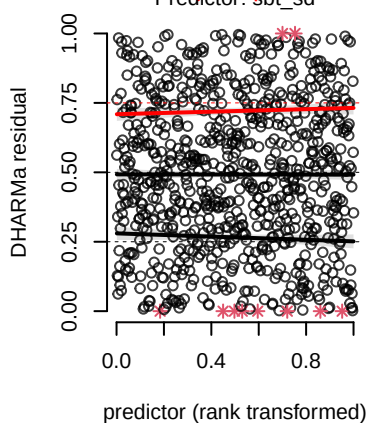
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



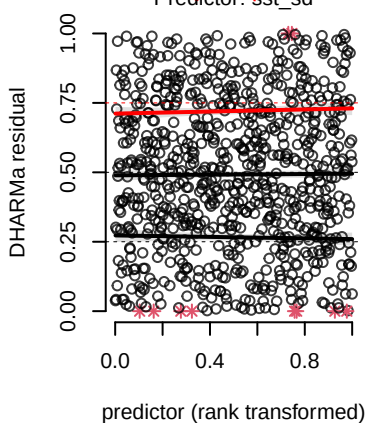
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



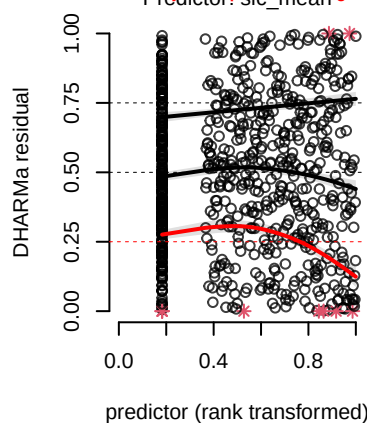
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



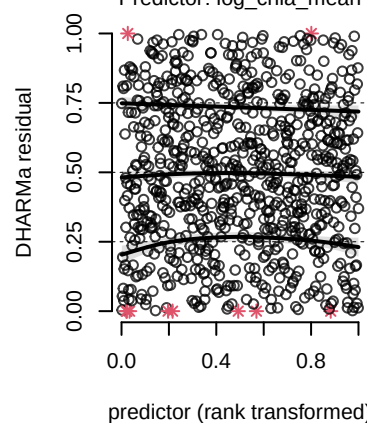
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_mean



DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.

